Three separate vertical tube holders

This is a first print prototype, not a verified final machine. Each holder supports one cap-down tube. Print one motorized gear station and two separate plain waiting holders. Move a tube from a waiting holder to the gear station to scan it. Each tube has its own resting place; the scan station is shared in turn.

Parts for three tubes

Use vertical_gear_module.scad to export these separate STL files:

OpenSCAD part | Quantity | Purpose
base | 1 | Stationary scan plate, motor holes and socket post
socket_gear | 1 | Holds the scanned tube cap and rotates around the post
motor_pinion | 1 | Small gear that fits the motor shaft
foot | 4 | Holds the scan motor clear of the table
waiting_holder | 2 | Separate unpowered stands for the other two tubes

Only one motor and driver are needed because one tube is scanned at a time. The other two motors you own can remain unused for this first build.

Measure before printing

1. Measure the outside diameter of the cap in millimeters. Set cap_diameter to that number. The default 16 mm is only a placeholder. The MRG challenge repository shows two tube sizes but provides no cap dimensions, so a physical ruler or caliper measurement is required.
2. Check the motor's two mounting holes, shaft diameter and flat-to-flat width. Change the four motor dimensions at the top of the SCAD file if yours differs. The values are starting points for a 28BYJ-48, not a verified fit for your motor.
3. Check that the shaft extends far enough past the 3 mm base to engage the 5 mm pinion. Check that the 30 mm feet clear the motor and its connector.
4. Make sure the tube cap is sealed before standing a filled tube cap down.

Export in OpenSCAD

1. Open vertical_gear_module.scad.
2. Change part to "motor_pinion". Press F6 (Render), then File > Export > Export as STL. Print this small piece first and check the fit on the unpowered motor shaft. Do not force it.
3. Repeat for "socket_gear"; check that the cap enters and lifts out.
4. Repeat for "base", "foot", and "waiting_holder". Print four feet and two waiting holders.

Bambu Studio starting settings

- Use a 0.4 mm nozzle, 0.20 mm layer height, 3 walls, and 15-20% gyroid infill.
- PLA is fine for a dry demonstration. PETG may be preferable if the holder will be cleaned often.
- Place the base flat, socket gear with its gear face on the plate, pinion flat, and feet upright. Inspect the sliced preview before printing.
- Supports should usually be unnecessary for these shapes. Add a brim if a narrow foot or pinion loses adhesion.
- Use Bambu Studio's Slice plate time as the real print estimate. As a planning range, allow roughly 1-2 hours for the gear station and 1-2 more hours for two waiting holders, depending on printer, material and final dimensions.

Assemble and test

1. Attach four feet under the base. Bolt the motor underneath with its shaft pointing up through the clearance hole.
2. Press the small pinion onto the unpowered motor shaft. Slip the socket gear over the stationary post. Turn both gears gently by hand; they should mesh without binding. A tiny amount of play is expected in this prototype.
3. Put one sealed tube cap in the socket. Confirm that it stays upright, its label clears the socket, and the base cannot tip. Fix the base to a larger board if needed.
4. Run the motor without a tube first, at low speed, then with one tube. Stop it fully for each photo. Put the finished tube back in its separate holder and transfer the next tube to the gear station. Do not leave it running unattended.

The gear teeth are simple prototype teeth. If they bind or skip, adjust the gear geometry before relying on the scanner. The driver board and power supply must be identified before providing electrical connections.